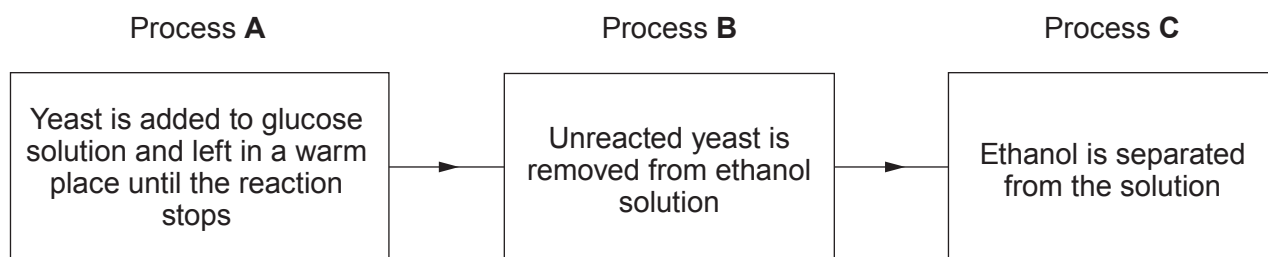
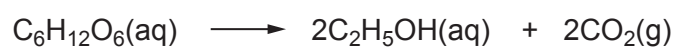


5. The chart describes the laboratory preparation of ethanol from glucose solution.



The equation for the reaction occurring in process A is shown below.



(a) (i) Give the name of the processes A, B and C.

[3]

A

B

C



(ii) A teacher wanted to show that ethanol is collected in process **C**.

- I. Tick (✓) the box next to the chemical test that the teacher would carry out to positively identify the liquid as ethanol. [1]

add bromine water

☐

add acidified potassium dichromate solution

☐

add silver nitrate solution

☐

add barium chloride solution

☐

- II. Tick (✓) the box next to the observation you would expect. [1]

orange to colourless

☐

orange to green

☐

green to orange

☐

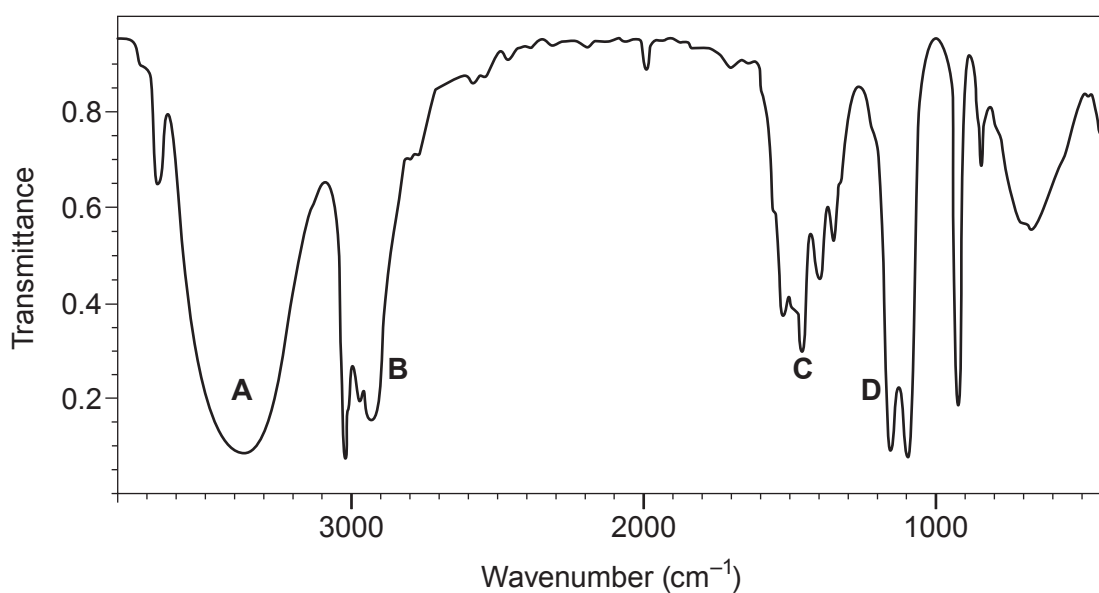
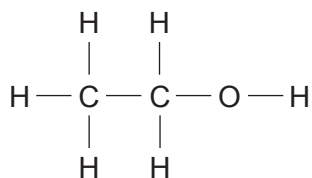
colourless to green

☐

(b) Alcohols can also be identified using infrared spectroscopy.

Bond	Wavenumber (cm^{-1})
$\text{C}=\text{C}$	1620 to 1670
$\text{C}=\text{O}$	1650 to 1750
$\text{C}-\text{H}$	2800 to 3100
$\text{O}-\text{H}$	2500 to 3550

(i) The structural formula and the infrared spectrum of ethanol are shown.



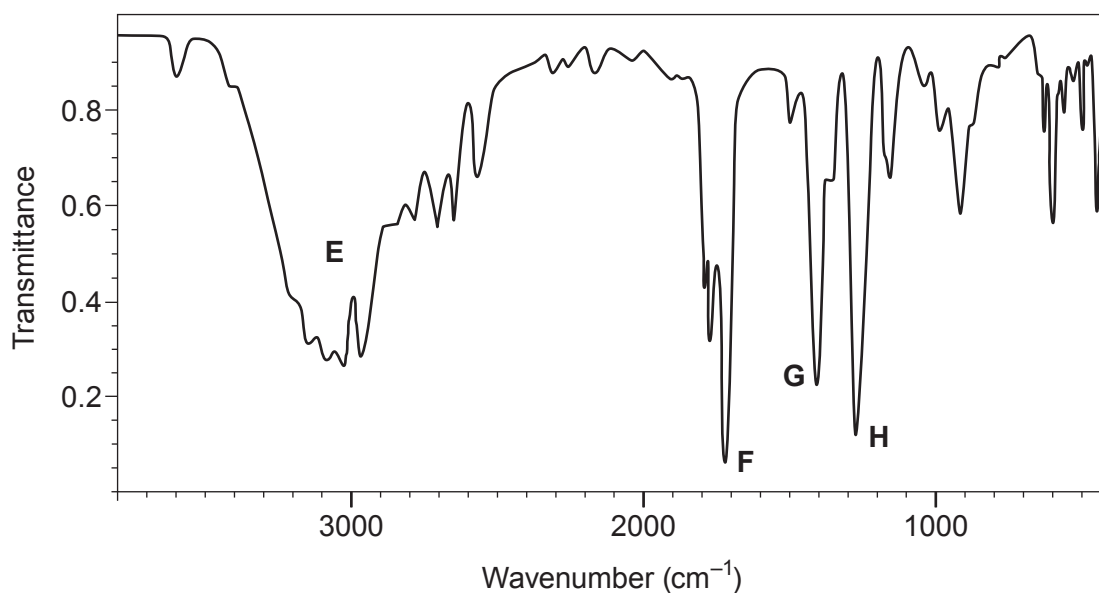
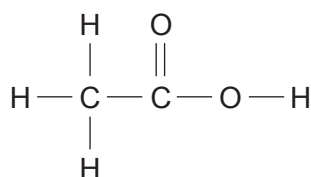
Give the **letter** of the peak that can be used to identify an alcohol.

[1]

Letter



- (ii) The structural formula and the infrared spectrum of ethanoic acid are shown.



Give the **letter** of the peak that can be used to distinguish ethanoic acid from ethanol.

[1]

Letter

- (c) Ethanol is found in alcoholic drinks.

- (i) Give **one** health problem associated with alcohol abuse over a **long** period of time.

[1]

.....

- (ii) Give **one** social problem associated with the excessive intake of alcohol (binge drinking) during an evening.

[1]

.....

